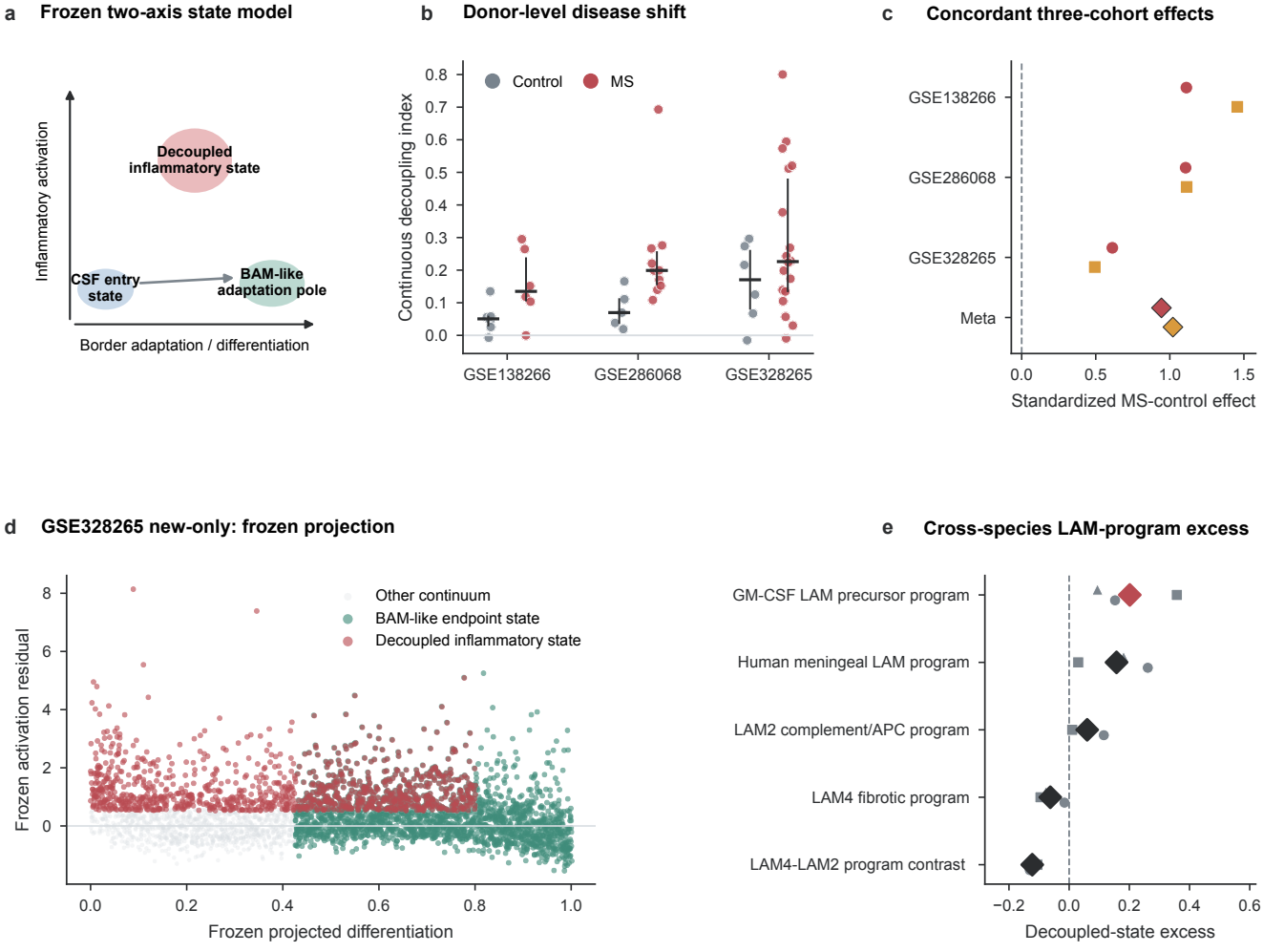

a
Frozen two-axis state model
Inflammatory activation
Decoupled inflammatory state
CSF entry state
BAM-like adaptation pole
Border adaptation / differentiation
b
Donor-level disease shift
Control
MS
Continuous decoupling index
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
GSE138266
GSE286068
GSE328265
c
Concordant three-cohort effects
GSE138266
GSE286068
GSE328265
Meta
0.0
0.5
1.0
1.5
Standardized MS-control effect
d
GSE328265 new-only: frozen projection
Other continuum
BAM-like endpoint state
Decoupled inflammatory state
Frozen activation residual
0
2
4
6
8
0.0
0.2
0.4
0.6
0.8
1.0
Frozen projected differentiation
e
Cross-species LAM-program excess
GM-CSF LAM precursor program
Human meningeal LAM program
LAM2 complement/APC program
LAM4 fibrotic program
LAM4-LAM2 program contrast
−0.2
0.0
0.2
0.4
0.6
Decoupled-state excess